

Table 1. Summary MGSM multilingual-transfer results for Qwen3-4B, Llama-3.2-3B-Instruct, Phi-4-mini-instruct, and Granite-3.3-2B-Instruct, macro-averaged equally across five languages. Values are full-sample means $\pm$ ten-group matched-question jackknife standard errors. Higher is better for every column.

Model	Method	Accuracy (%) $\uparrow$	Spanish relevance (0–2) $\uparrow$	Instruction relevance (0–2) $\uparrow$	Fluency (0–2) $\uparrow$
Qwen3-4B	Original	41.6 $\pm$ 2.5	0.00 $\pm$ 0.00	1.76 $\pm$ 0.03	1.47 $\pm$ 0.03
	A-LQR	22.4 $\pm$ 2.3	<u>1.81 $\pm$ 0.02</u>	1.51 $\pm$ 0.04	1.08 $\pm$ 0.02
	H_∞ (ours)	32.2 $\pm$ 2.4	1.44 $\pm$ 0.03	1.66 $\pm$ 0.03	1.25 $\pm$ 0.04
Llama-3.2-3B	Original	33.0 $\pm$ 1.6	0.00 $\pm$ 0.00	1.68 $\pm$ 0.03	1.44 $\pm$ 0.03
	A-LQR	<u>27.6 $\pm$ 2.6</u>	<u>1.50 $\pm$ 0.03</u>	<u>1.57 $\pm$ 0.04</u>	<u>1.17 $\pm$ 0.04</u>
	H_∞ (ours)	12.2 $\pm$ 2.4	1.47 $\pm$ 0.02	1.02 $\pm$ 0.05	0.91 $\pm$ 0.04
Phi-4-mini	Original	45.4 $\pm$ 2.6	0.00 $\pm$ 0.00	1.79 $\pm$ 0.02	1.52 $\pm$ 0.04
	A-LQR	0.0 $\pm$ 0.0	1.09 $\pm$ 0.04	0.04 $\pm$ 0.01	0.04 $\pm$ 0.01
	H_∞ (ours)	35.2 $\pm$ 2.6	1.21 $\pm$ 0.04	1.56 $\pm$ 0.04	1.05 $\pm$ 0.03
Granite-3.3-2B	Original	22.2 $\pm$ 2.0	0.00 $\pm$ 0.00	1.34 $\pm$ 0.03	1.18 $\pm$ 0.03
	A-LQR	18.8 $\pm$ 2.0	<u>0.04 $\pm$ 0.02</u>	1.04 $\pm$ 0.03	0.80 $\pm$ 0.03
	H_∞ (ours)	21.2 $\pm$ 1.9	0.00 $\pm$ 0.00	1.33 $\pm$ 0.03	1.18 $\pm$ 0.03